(A) 1. Tranzistorul NMOS.

a) Prezentare, funcționare, avantaje/dezavantaje față de tr. bipolar

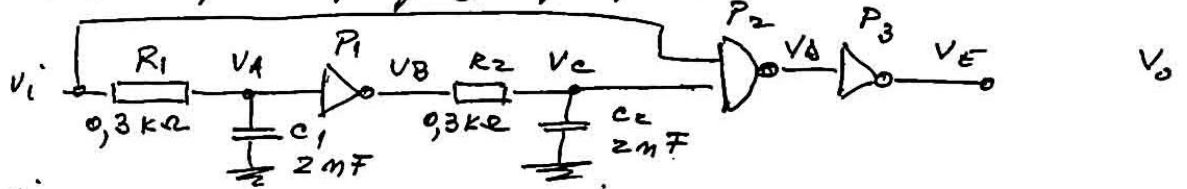
b) Regimurile de funcționare

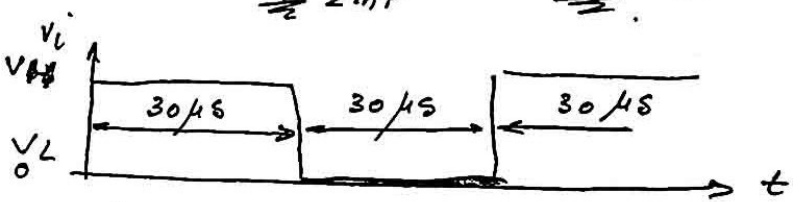
c) Să se ridice caracteristica de intrare și ieșire a unui tranz. NMOS având următoarele date: $V_T = 4^V$; $V_{DD} = 10^V$; $\frac{W}{L} = 1$; $k = 100\,\mu A/V^2$

$V_i = 0 \div 10^V$

2. a) Se dă circ. din fig. de mai jos. Se cere să se ridice diagramele de timp de funcționare în punctele A, B, C, D, E.

b) Să se determine valoarea timpilor rezultați la ieșire. Se vor neglija timpii de propagare pe porți.





$V_{DD} = 5^V$; $V_T = 1,5^V$, $V_H = 3,5^V$; $V_L = 0,2^V$

SUBIECT A

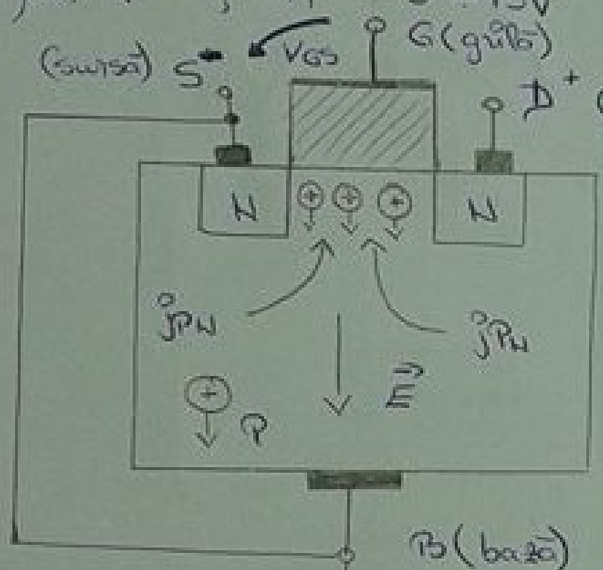
A1. Tranzistorul NMOS

a) Prezentare, funcționare, avantaje/dezavantaje față de tr. bipolar

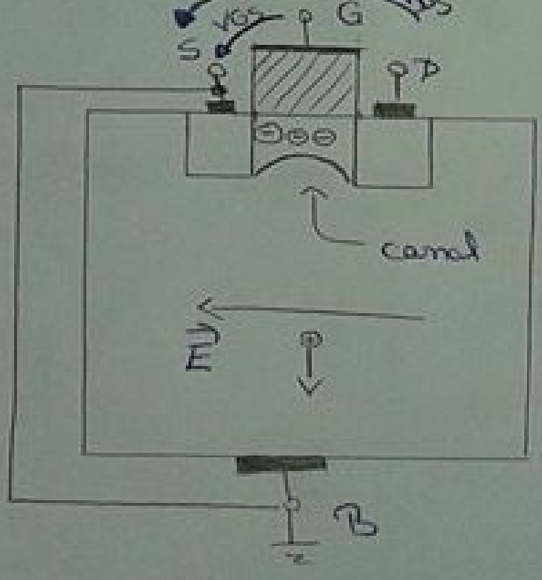
b) Regimuri de funcționare

c) Să se ridice caracteristica de intrare și ieșire a unui tranzistor NMOS având datele: $V_T = 4V$; $V_{DD} = 10V$; $\frac{W}{L} = 1$; $K = 100\ \mu A/V^2$; $V_I = 0 \div 10V$

a)



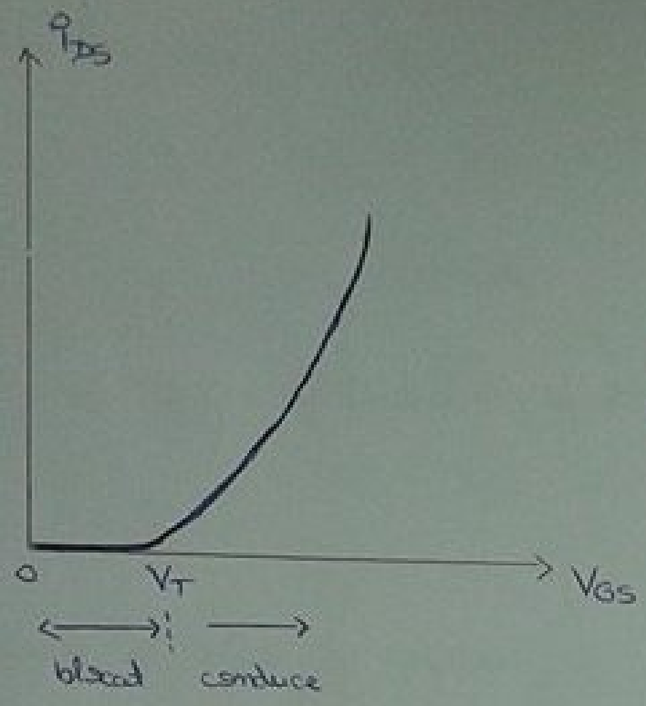
* MOS - metal, SiO_2, semiconductor
* are la deplasarea unui singur tip de sarcini



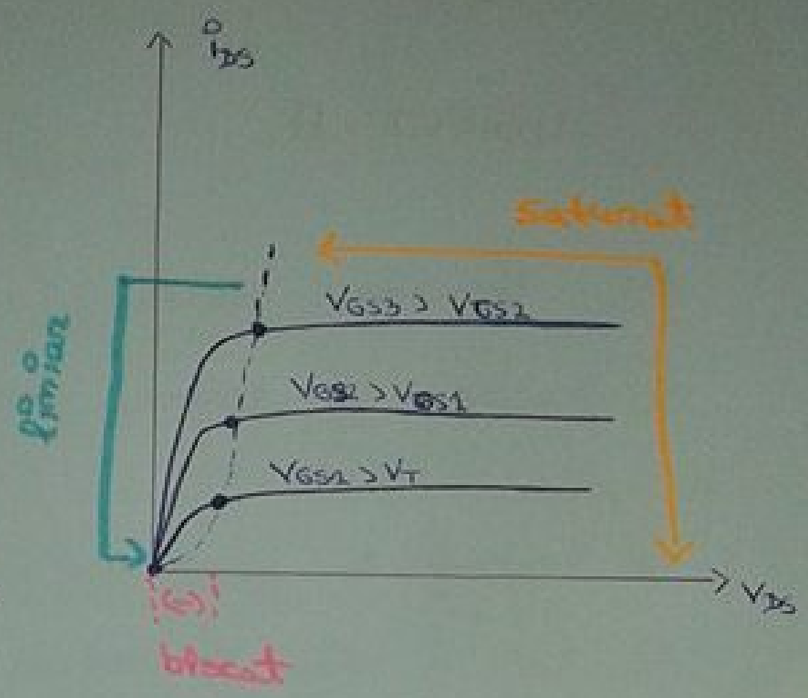
$V_{DS} > 0\ V$

$V_{GS} > 0\ V$

* MOS cu canal indus

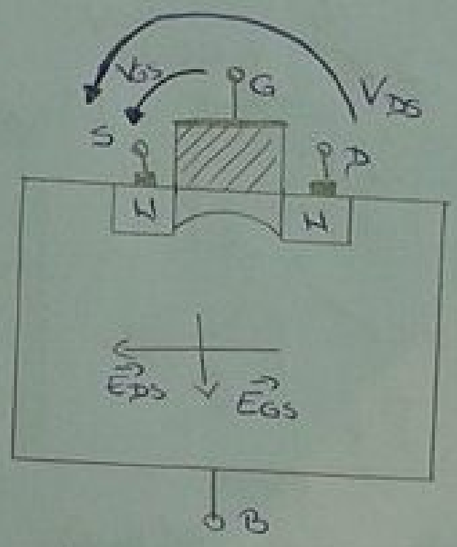


* caracteristica de intrare



* caracteristica de ieșire

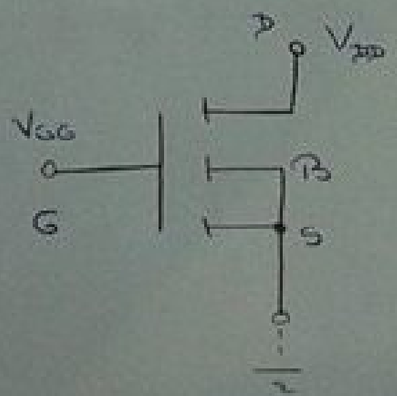
$$V_{DS} = V_{GS} - V_T$$



* tranzistor MOS: ... se formează în funcție de:

~ cu canal indus

~ cu canal inițial (influențează modul de comandare)



- Regim BLOCAT

$V_{GS} < V_T \Rightarrow I_{DS} = 0$

- Regim LINIAR

$V_{GS} > V_T$

$V_{DS} < V_{GS} - V_T$

$$I_{DS} = 2K'\left[(V_{GS} - V_T) \cdot V_{DS} - \frac{V_{DS}^2}{2}\right]$$

$K = K' \cdot \frac{W}{L}$; K' - factor de conductanță

W - lungimea canalului

L - lățimea canalului

$K' = [\mu A/V^2]$

- Regim SATURAT

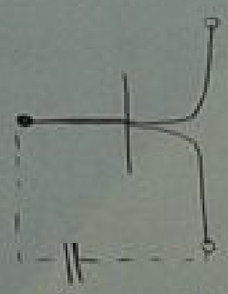
$V_{GS} \geq V_T$

$V_{DS} \geq V_{GS} - V_T$

$$I_{DS} = K(V_{GS} - V_T)^2$$

<u>Bipolar</u>

- pe insula izolatoare se pune tranzistorul

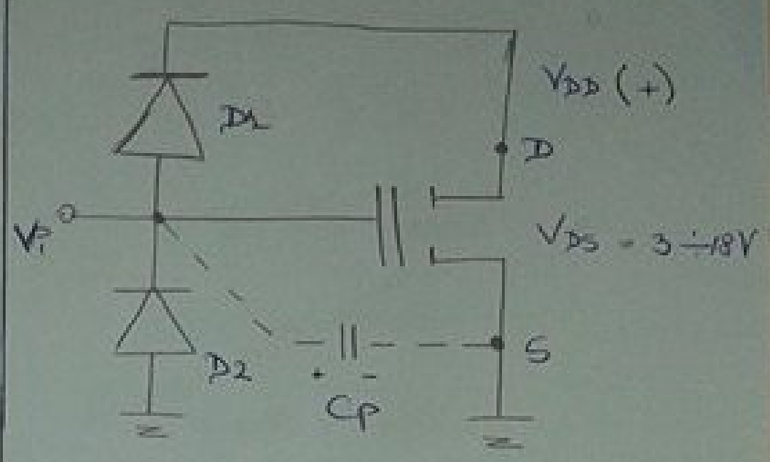


$C_P \rightarrow$ datorită joncțiunii

<u>MOS</u>

- mai ieftin
- poate juca rolul unui rezistor
- rezistența canalului e destul de mare: $\tau = RC$
- consumul cel mai mic
- implică regim special de utilizare

- M are pe undă să se descarce



T → conduce ⇒ $R_{GS} \to \infty$ ⇒ Cp nu are pe unde să se descarce

* toate circuitele numerice au la intrare circuite de protecție,
* Cp poate ajuta la memorare (reținerea unei valori binare)

c)

$$\left.\begin{array}{l} V_{DS} = V_{GS} - V_T \\ \text{Dar } V_{DS} = V_i^o \end{array}\right\} \Rightarrow V_i^o = V_{GS} - V_T \Rightarrow V_{GS} = V_i^o + V_T$$

$$V_{GS} = 10V + 4V = 14V$$

$V_{GS} = 14V > 4V = V_T \Rightarrow$ regim liniar

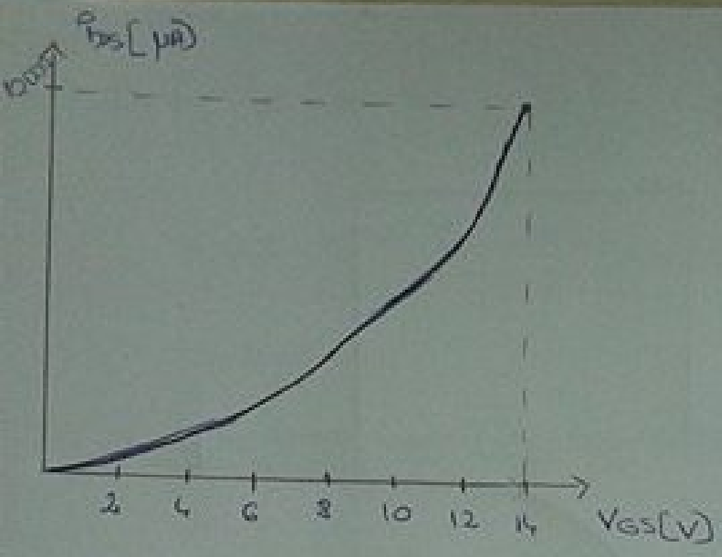
$$I_{DS} = 2K_n\left[(V_{GS} - V_T)\cdot V_{DS} - \frac{V_{DS}^2}{2}\right]$$

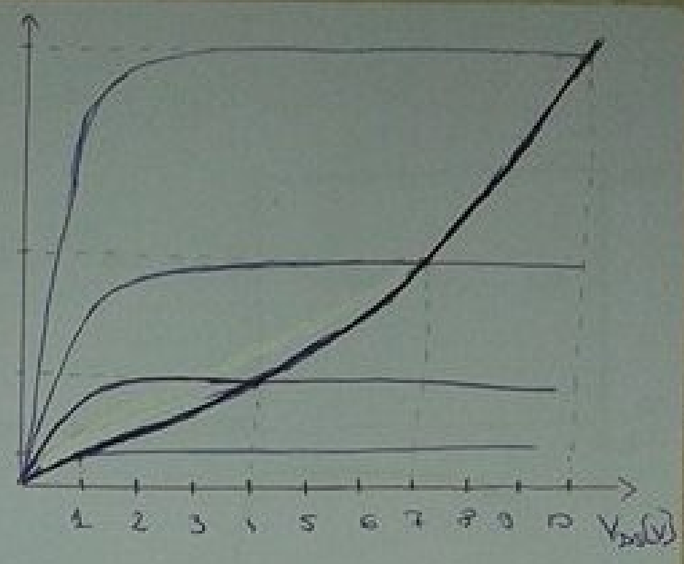
$$I_{DS} = 2\cdot 100\cdot 10^{-6}\left[10\cdot 10 - \frac{100}{2}\right] =$$

$$= 2\cdot 10^2\cdot 10^{-6}\cdot 50 =$$

$$= 10.000\cdot 10^{-6} =$$

$$= 10.000\ \mu A$$





$V_{DS} = V_{GS} - V_T$

- $V_{GS} = 5V$

  $i_{DS} = 100\ \mu A$

  $V_{DS} = 5V - 4V = 1V$

- $V_{GS} = 8V$

  $i_{DS} = 1600\ \mu A$

  $V_{DS} = 8V - 4V = 4V$

- $V_{GS} = 11V$

  $i_{DS} = 4900\ \mu A$

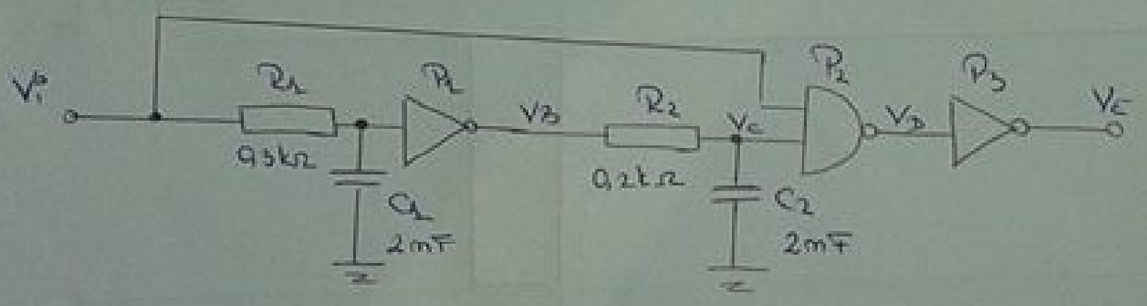
  $V_{DS} = 11V - 4V = 7V$

- $V_{GS} = 14V$

  $i_{DS} = 10.000\ \mu A$

  $V_{DS} = 14V - 4V = 10V$

A 2.



$V_{CC} = 5V$

$V_T = 1,5V$

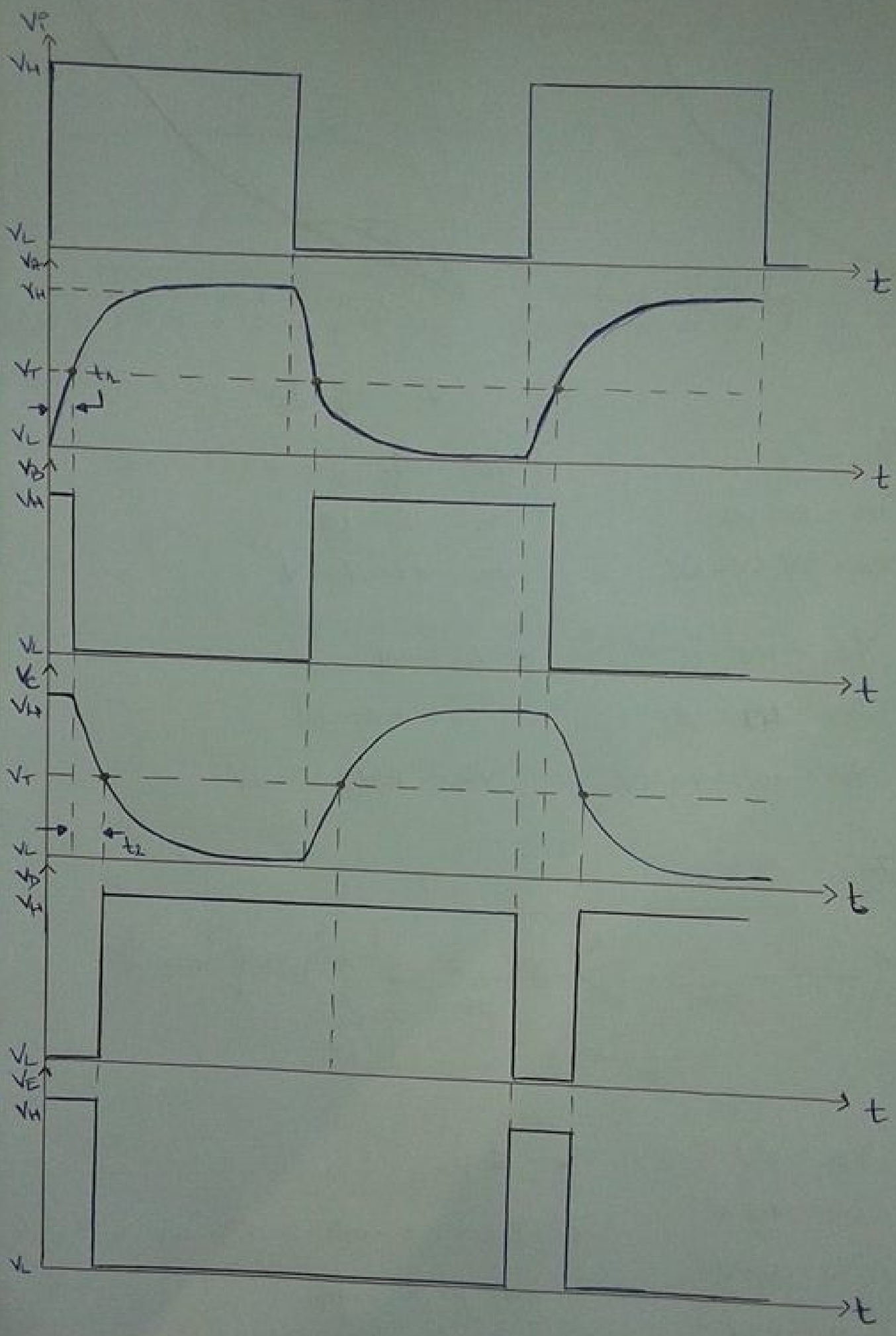
$V_H = 3,5V$

$V_L = 0,2V$

$T_a = 60\ \mu s$

$\tau = R \cdot C = 0,3 \cdot 10^3 \cdot 2 \cdot 10^{-9} = 0,6 \cdot 10^{-6} =$

$= 0,6\ \mu s < 30\ \mu s$


V_i^o
V_H
V_L
t
V_A
V_H
V_T
t_1
V_L
t
V_B
V_H
V_L
t
V_E
V_H
V_T
t_2
V_L
t
V_D
V_H
V_L
t
V_E
V_H
V_L
t

$$t_1 = R_1 \cdot C_1 \cdot \ln \frac{V_H - V_L}{V_H - V_T} = 0{,}3 \cdot 10^3 \cdot 2 \cdot 10^{-9} \ln \frac{3{,}5 - 0{,}2}{3{,}5 - 1{,}5} =$$

$$= 0{,}6 \cdot 10^{-6} \ln \frac{3{,}3}{2} = 0{,}6 \cdot 10^{-6} \ln 1{,}65 = 0{,}6 \cdot 0{,}5 \cdot 10^{-6} =$$

$$= 0{,}3\,\mu s$$

$$t_2 = R_2 \cdot C_2 \cdot \ln \frac{V_L - V_H}{V_L - V_T} = 0{,}2 \cdot 10^3 \cdot 2 \cdot 10^{-9} \ln \frac{0{,}2 - 3{,}5}{0{,}2 - 1{,}5} =$$

$$= 0{,}4 \cdot 10^{-6} \ln \frac{3{,}3}{1{,}3} = 0{,}4 \cdot 10^{-6} \cdot 0{,}93 = 0{,}37\,\mu s$$

$$t_g = t_1 + t_2 = 0{,}67\,\mu s$$

$$\Rightarrow T_1 = 0{,}67\,\mu s$$

$$T_2 = 60\,\mu s - 0{,}67\,\mu s = 59{,}33\,\mu s$$